Ashwin Chembu

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PROFILE

Applied-AI engineer building retrieval-augmented products with document ingestion, vector search, evidence extraction, and cited answers. Connects grounded AI workflows to usable interfaces and backend services.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL

AI / Retrieval: RAG, LangChain, OpenAI, Azure OpenAI, Pinecone, FAISS

Mechanisms: Document ingestion, vector embeddings, intent routing, evidence extraction, citations

Product / Backend: React, FastAPI, Node.js, Express, MongoDB, Redis, AWS S3

Validation: pytest, grounded answers, generator/critic refinement, human review

EXPERIENCE

Business Technology Solutions Associate | ZS Associates **South San Francisco, CA | Jun 2024-Present**

- Built React/**FastAPI RAG** assistants with AWS document ingestion and Redis services for enterprise knowledge search.
- Contributed **FastAPI**/Uvicorn model workflows and pytest checks to validate enterprise **RAG** services and AI behavior.
- Built Python/Flask endpoints connecting AWS S3 and Smartsheet APIs to automate document storage and cloud handoffs.
- Translated stakeholder needs into testable rules, API integrations, source-to-target mappings, and delivery documents.

Founder and Lead Developer | Tallyrus **Jun 2023-Sep 2025**

- Built Python/**LangChain** essay-feedback workflows with **Pinecone** and OpenAI for retrieval-augmented educator review.
- Built Flask services with MongoDB persistence and S3 storage to manage documents and support essay-feedback flows.
- Designed React rubric interfaces and feedback loops so educators could review essays with consistent scoring logic.

Data Analyst / Web Development Lead | Tech4Good at UCSC **Mar 2023-Jun 2025**

- Built TypeScript/Figma API chat, polling, and export workflows in a collaborator-owned repository for classroom collaboration.
- Implemented 20+ attributed Node/Express routes with MongoDB persistence and logging for chat, text entry, and polls.
- Analyzed **43,340** interaction rows across **2,805** sessions with Python/Pandas and built Excel/XLSX exports for research.

PROJECTS

Lume | AI Meeting Intelligence **Dec 2025-Jan 2026**

- Designed **RAG** with vector retrieval, intent routing, evidence extraction, and generator/critic passes for cited answers.
- Built reusable Project Brain knowledge bases retaining transcript/file terminology and relationships for grounded Q&A.
- Built React and Node/Express flows with MongoDB/S3 to capture, store, search, and query meetings and project files.
- Connected meeting capture, document search, and reusable context through responsive React and WebSocket updates.

Azure RAG | Document Search Proof of Concept **Public proof of concept**

- Built Python document search with Azure OpenAI embeddings, **LangChain,** and **FAISS** to retrieve relevant source passages.
- Added PDF/TXT/Markdown ingestion, chunking, vector indexing, and source citations to ground interactive document Q&A.
- Documented Azure embedding/chat-model setup and a 30+ document corpus for repeatable retrieval and grounded Q&A.

Private Filedrop | Secure File Exchange **Jun 2026-Aug 2026**

- Built password-protected uploads and API-key-protected Express APIs with OpenAPI docs for secure file exchange.
- Added local/S3-compatible storage, migration tooling, version history, and unguessable links for durable file handoffs.
- Exposed MCP file tools with HMAC-SHA256 webhooks and timing-safe credential checks for signed service events.

EDUCATION

University of California, Davis | **B.S. Computer Science** | **GPA: 3.7** Graduated Jun 2025

CERTIFICATIONS

AWS Certified Cloud Practitioner | Google Data Analytics Certificate

LINKS

Portfolio: https://ashchembu.com | LinkedIn: linkedin.com/in/ashwin-chembu-a5292b205 | GitHub: github.com/ashwinchembu